

AB RENEWABLE ENERGY ENGINEERING

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FUEL AND COMBUSTION

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1. The substance that carries energy and produces power when ignited.

- a. Heat
- b. Fuel
- c. Air
- d. None of the above

2. The elements necessary for combustion.

- a. Fire
- b. Air
- c. Fuel
- d. All of the above
- e. Any two of the above

3. The air required for complete combustion of combustible gases after solid and/or liquid fuels are burned.

- a. Stoichiometric air
- b. Primary air
- c. Secondary air
- d. All of the above

4. The excessive amount of tar produced during the combustion of agricultural wastes is primarily due to ____.

- a. insufficiency of fuel
- b. less air used
- c. too much air used
- d. None of the above

5. During gasification of biomass fuel, the primary gas produced is ____.

- a. carbon monoxide
- b. carbon dioxide
- c. methane and carbon dioxide
- d. All of the above

CO

6. The gas produced during perfect combustion of fuel.

- a. Carbon monoxide
- b. Carbon dioxide
- c. Hydrogen
- d. None of the above

CO₂

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7. The gas produced from partial combustion of agricultural wastes which is odorless, tasteless, colorless, and flammable.

- a. Biogas
- b. CO
- c. CH₄
- d. None of the above

8. The bulk density of rice husks.

- a. 50-80 kg/m³
- b. 100-120 kg/m³
- c. 140-160 kg/m³
- d. None of the above

9. The air requirement for gasification of agricultural wastes ranges from ____.

- a. 10-20 % of stoichiometric air (SA)
- b. 20-30 % of SA
- c. 20-40 % of SA
- d. None of the above

10. The caloric value of fuel is ____.

- a. the amount of heat contained by the fuel per unit weight
- b. the weight of fuel to be heated
- c. the heat needed to burn a certain weight of fuel
- d. None of the above

11. The amount of heat required to raise the temperature of one pound water by one degree Fahrenheit.

- a. kcal — metric
- b. BTU — SI
- c. Joule
- d. None of the above

12. Pyrolysis requires around ____ of the stoichiometric air of fuel.

- a. 10-20 %
- b. 20-30 %
- c. 30-40 %
- d. None of the above

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13. Rice husk fuel will be used as a replacement for LPG fuel in providing heat to a poultry farm. If two 50-kg LPG tanks are needed per day (24 hours) operation, what would be the amount of rice husk fuel in kilogram that will be consumed per hour? Assume an LPG heating value of 11,000 kcal/kg and a conversion efficiency for the system of 60%. Heating value of rice husk is 3,000 kcal/kg.

- a. 15 kg per hour
- b. 25 kg per hour
- c. 36 kg per hour
- d. None of the above

14. The calorific value of diesel.

- a. 3,000 to 4,000 kcal/kg
- b. 4,000 to 5,000 kcal/kg
- c. 6,000 to 7,000 kcal/kg
- d. 8,000 to 9,000 kcal/kg
- e. 10,000 to 11,000 kcal/kg
- f. 11,000 to 12,000 kcal/kg
- g. None of the above

15. The calorific value of LPG.

- a. 3,000 to 4,000 kcal/kg
- b. 4,000 to 5,000 kcal/kg
- c. 6,000 to 7,000 kcal/kg
- d. 8,000 to 9,000 kcal/kg
- e. 10,000 to 11,000 kcal/kg
- f. 11,000 to 12,000 kcal/kg
- g. None of the above

16. The calorific value of coal.

- a. 3,000 to 4,000 kcal/kg
- b. 4,000 to 5,000 kcal/kg
- c. 6,000 to 7,000 kcal/kg
- d. 8,000 to 9,000 kcal/kg
- e. 10,000 to 11,000 kcal/kg
- f. 11,000 to 12,000 kcal/kg
- g. None of the above

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17. The calorific value of agri-residues.

- a. 3,000 to 4,000 kcal/kg
- b. 4,000 to 5,000 kcal/kg
- c. 6,000 to 7,000 kcal/kg
- d. 8,000 to 9,000 kcal/kg
- e. None of the above

18. The calorific value of gasoline.

- a. 3,000 to 4,000 kcal/kg
- b. 4,000 to 5,000 kcal/kg
- c. 6,000 to 7,000 kcal/kg
- d. 8,000 to 9,000 kcal/kg
- e. 10,000 to 11,000 kcal/kg
- f. 11,000 to 12,000 kcal/kg
- g. None of the above

19. A sample analysis that gives information about the fixed carbon, volatile matter and ash content of fuel.

- a. Calorific analysis
- b. Proximate analysis
- c. Ultimate analysis
- d. All of the above

20. Ultimate analysis of coal shows that carbon content is 81.3% by weight while hydrogen, oxygen and sulphur contents are 5.3, 9.8, and 1.9%, respectively. What is the stoichiometric air of coal?

- a. 4.7 kg air per kg fuel
- b. 10.9 kg air per kg fuel
- c. 15.3 kg air per kg fuel
- d. None of the above

21. A high-cetane-fueled engine uses ____.

- a. diesel as fuel and is difficult to start at lower temperature
- b. diesel as fuel and is easy to start at lower temperature
- c. gasoline fuel and is easy to start at lower temperature
- d. None of the above

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22. What is the heating value of fuel with ultimate analysis result as follows: carbon - 38.9%; hydrogen - 4.81%; sulfur - 0.10%; and oxygen - 34.4%?

- a. 10.3 MJ/kg
- b. 13.9 MJ/kg
- c. 15.4 MJ/kg
- d. None of the above

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GOD BLESS !!!

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